What Happens Next? Closed- and Open-World Video Action Recognition on a 33-Class Something-Something V2 Subset

Florian Guillaumey
École Polytechnique — Team 8
florian.guillaumey@polytechnique.edu

Andrea Signoretti
École Polytechnique — Team 8
andrea.signoretti@polytechnique.edu

May 27, 2026

Abstract

*We present a two-track study of temporal action prediction on a 33-class subset of Something-Something V2 [3], where only the first ≈40% of each clip (four JPEG frames) is given. In **Track A (Closed World)**, all models are trained from scratch. We show that two data-centric decisions dominate architecture: a regularisation warm-up (+36.3 pp) and matching the model's frame count to the real clip length (+4.4 pp, beating a 2× capacity scale-up). A five-model log-softmax ensemble combining two TSM architectures (ResNet18 and ResNet50), one VideoFormer-Lite, and one CE-trained TSM ablation reaches **46.75%** top-1 on the official validation set; the best single model (TSM-ResNet50, rotating folds) reaches **42.85%**. In **Track B (Open World)**, we fine-tune V-JEPA 2 [1] and VideoMAE [12] with progressive unfreezing and layer-wise learning-rate decay, reaching **0.6586** Kaggle top-1 with a 14-model ensemble (16th place in both tracks).*

1. Problem Description

Task. The challenge classifies four-frame JPEG clips into 33 motion-defined action classes (e.g. *Pouring something into something*, *Pulling from left to right*). Each clip contains only the *anticipatory* portion of the source SSv2 video (shipped `frame_003` sits at median frame fraction 0.40), so the outcome is deliberately withheld. The task thus requires reasoning about ongoing motion, not recognising a final state [3].

Dataset. The split provides 44,993 training, 6,745 validation and 6,913 test clips over 33 classes. Classes are heavily imbalanced: the largest has 3,170 examples and the smallest only 162 (ratio ≈ 20×). The KL divergence between train and validation class distributions is 0.088, confirming the splits are representative.

Evaluation and tracks. Kaggle scores top-1 accuracy on public+private leaderboard. **Track A (Closed World)** prohibits pretrained models and external data. **Track B (Open World)** permits pretrained backbones and external corpora but not future frames or test labels. The task is fundamentally hard for three reasons: (i) the labels are motion-defined, not appearance-defined; (ii) four frames is far fewer than the 8–32 used in standard video recognition; (iii) the decisive cue—the action's outcome—is withheld by design.

2. Methods

2.1. Track A — Closed World

Architecture overview. We implemented and compared three temporal architectures, all built on a ResNet backbone [4] and accepting input clips of shape (B, T, C, H, W) with $T{=}4$, $H{=}W{=}224$.

CNNBaseline. A ResNet18 processes each frame independently, and the T frame-level feature vectors (each of size 512) are averaged before the classifier head. This gives a 0-parameter temporal operation but captures no inter-frame dynamics. Total: ≈ 11.2 M parameters.

TSM-ResNet (Temporal Shift Module) [7]. The key idea is to inject temporal information *before* each residual block, while all spatial features are still present at full resolution. Concretely, for a feature map $\mathbf{x} \in \mathbb{R}^{B\times T\times C\times H\times W}$, the TSM operation shifts a fraction 1/fold_div of channels along the time axis before each convolution:

$$\text{TSM}(\mathbf{x})_{t,c} = \begin{cases} \mathbf{x}_{t-1,c} & c < C/4 \quad \text{(backward shift)} \\ \mathbf{x}_{t+1,c} & C/4 \le c < C/2 \quad \text{(forward shift)} \\ \mathbf{x}_{t,c} & \text{otherwise} \end{cases} \tag{1}$$

This adds **zero parameters and zero FLOPs**: temporal mixing is free. After 8 residual blocks (each preceded by TSM), a global average pool produces one 512-d vector per frame; the T vectors are then averaged and a linear head (512×33) produces class logits.

We ran two backbone variants:

- **TSM-ResNet18**: ResNet18 backbone [4], ≈ 11.2 M parameters total.
- **TSM-ResNet50**: ResNet50 backbone, ≈ 23.6 M parameters (of which 23.5 M in the backbone, 0.07 M in the head). The head is now 2048×33 due to ResNet50's wider final layer.

Both add zero parameters from TSM; the full parameter count comes from the ResNet backbone and the linear head alone.

VideoFormer-Lite (VFL). VFL separates spatial and temporal processing into two explicit stages (Fig. 1). *Stage 1 (spatial):* ResNet18 processes each frame independently, producing T feature vectors of size d=512 after global average pooling. *Stage 2 (temporal):* A learnable `[CLS]` token is appended, giving a sequence of T+1=5 tokens. A stack of n_ℓ pre-norm Transformer blocks [14] then performs global self-attention over all frames simultaneously. Each block contains multi-head attention (8 heads) and an MLP (expansion ratio r=4), and its parameter count is:

$$P_{\text{block}} = \underbrace{4d^2}_{\text{Q,K,V,O proj.}} + \underbrace{2\,r\,d^2}_{\text{MLP}} + \underbrace{4d}_{\text{layer norms}} \approx (4+2r)\,d^2 \quad (2)$$

With d=512, r=4: $P_{\text{block}} \approx 3.1$ M. For n_ℓ=3 blocks the Transformer adds ≈9.4 M parameters on top of the ≈11.2 M ResNet18 backbone, for a total of ≈20.7 M.

Where temporal mixing happens — the key difference. TSM mixes time at every spatial resolution, inside every residual block, *before* any spatial pooling. After 8 blocks of depth, each frame's representation already encodes a receptive field spanning all T frames. VFL instead pools all spatial information first, then applies global attention over the T+1 pooled vectors. This means VFL sees motion only at the level of frame-averaged features, while TSM sees it at each (h, w) location. This distinction matters for SSv2: motion cues like a finger moving or an object edge displacing live in local spatial regions that are lost after pooling.

Critical frame-count fix (+4.4 **pp).** The clips contain exactly four real frames. Setting `num_frames` > 4 triggers linspace interpolation in the dataset loader, which silently duplicates frames. For TSM, duplicated frames produce two consecutive tokens with identical content; the shift in Eq. (1) then reads the same value twice, contributing zero temporal gradient for those channel positions. Fixing `num_frames` = 4 improved internal val by +4.41 pp (53.32→57.73%) and `val_dir` by +3.34 pp. Scaling to ResNet50 at T=8 with a fixed split *regressed* to 45.26% internal (Fig. 3), confirming that corrupted temporal input cannot be fixed by larger capacity. After correcting this, a TSM-ResNet50 at T=4 with rotating folds reaches **42.85%** `val_dir`—the new best single model (Fig. 2).

Technique	Setting	Source
Reg. warm-up	10 ep, no MixUp/CutMix	**this work**
MixUp / CutMix	α=0.2 / p=0.25, after ep 10	[15, 16]
Focal loss	γ=2	[8]
Label smoothing	ε=0.1	[11]
RandAugment	2 ops, mag. 0.5	[2]
Stochastic depth	DropPath 0.1	[5]
Optimiser	AdamW, η=10^{-3}, cosine WR	[10]
Horizontal flip	**disabled**	label semantics

Table 1. Track A training recipe. Reg. warm-up is our contribution; all other components are standard practices from the cited works.

Regularisation recipe. The most critical element is a *warm-up delay* on MixUp [16] and CutMix [15]: both are disabled for the first 10 epochs. Without it, VFL stalls at 9.45% (killed at epoch 4) vs. 45.78% with it (+36.3 pp). The mechanism is straightforward: MixUp creates interpolated labels (e.g. 60% class A, 40% class B) that a randomly initialised network cannot resolve, preventing basic feature formation. The remaining stack (Table 1) includes focal loss [8] (γ=2, concentrating gradient on confusable pairs), label smoothing, RandAugment, stochastic depth and AdamW with cosine warm restarts [9]. Horizontal flip is disabled: it would invert direction-encoded class labels (e.g. left→right becomes right→left).

Rotating folds. Instead of a fixed 80/20 train/val split, we train with a rotating validation fold: each epoch the held-out fold shifts by one index (5 folds, so fold$_t$ = t mod 5). This ensures every sample participates in 100% of training epochs rather than 80%, with no fixed validation oracle. Model selection uses the rolling average of the last 5 validation accuracies (one complete rotation). This adds +1.2 pp for TSM and +4.3 pp for VFL; VFL benefits more because its global-attention head is data-hungry and lacks the structural temporal prior that TSM's shift provides.

Track A ensemble. The best five-model ensemble fuses checkpoints by log-softmax late fusion. Three models use 10-crop TTA (5 spatial crops × 2 flips): `TSM-ResNet18-v2`, its rotating-fold variant, and `VFL-Ultra-rotating`. Two run without TTA: `TSM-ResNet50-rotating` and a CE-trained TSM ablation model. Averaging in log-probability space downweights over-confident wrong predictions more than raw logit averaging does. The ensemble reaches **46.75%** on `val_dir` (Fig. 6).

2.2. Track B — Open World

We selected two self-supervised families: **VideoMAE** [12] (masked-autoencoder pre-training on Kinetics-400 [6]) and **V-JEPA 2** [1], which reconstructs masked regions in *feature* space rather than pixel space, using the self-supervised `vjepa2-vitl-fpc64-256` checkpoint without any SSv2 labels (integrity requirement, Sec. 3.4).

Progressive unfreezing with LLRD. We fine-tune V-JEPA in three phases with layer-wise LR decay (LLRD), where block ℓ receives $\eta \cdot \lambda^{\ell}$ (λ=0.75). Phase 0 (frozen backbone, head only): 0.672 EMA internal val. Phase 1 (top-6 blocks unfrozen): 0.705. Phase 2 (full encoder): 0.728 internal, 0.6411 `val_dir`.

Window-capped source-frame re-extraction. V-JEPA benefits from 16 input frames. We re-extract 16 frames per clip from the source WebM files, hard-bounded to $[0, \text{window\_end}]$ where `window_end` is located by matching `frame_003` against source frames by pixel-level MAE, and capped at 50% of the source clip so the withheld outcome is unreachable.

Pseudo-labelling. Soft labels from the Phase-2 ensemble are used for test clips with max confidence $\geq$ 0.85; V-JEPA-pseudo reaches 0.6410 Kaggle. The growing `val_dir`→Kaggle gap warned against further iterations.

Cross-preprocessing ensemble. The final 14 checkpoints come from three families with distinct preprocessing chains (V-JEPA: 16-frame 256 px; VideoMAE: 4-frame 224 px; TSM-ResNet50: 4-frame). Uniform averaging of softmax tensors reaches 0.6586.

2.3. Reused vs. own code

We *reuse*: the course starter repository (Hydra configuration, `VideoFrameDataset`, `train.py` skeleton); torchvision ResNet and R(2+1)D backbones; HuggingFace V-JEPA 2 and VideoMAE checkpoints (Track B). Our *own contributions*: the TSM block wrapper (reimplemented from Lin et al. [7]); VideoFormer-Lite; the regularisation warm-up recipe; rotating k-fold training; log-softmax ensemble and 10-crop TTA pipeline; and the full Track B stack (window-capped re-extraction, LLRD fine-tuning, pseudo-labelling loop).

3. Results

Overall. Table 2 reports headline results. Track A reaches **46.75%** on `val_dir` from scratch; Track B reaches **0.6586** Kaggle top-1 with the 14-model uniform ensemble. The gap between tracks (≈20 pp) quantifies the practical value of large-scale pretraining unavailable in the closed track.

Trk	Method	val dir	Kaggle
A	TSM-R18, single best	41.68%	—
A	TSM-R50, rotating (no TTA)	42.85%	—
A	**5-model ens.**	**46.75%**	—
B	V-JEPA-ft standalone	64.11%	0.6361
B	V-JEPA-pseudo	65.19%	0.6410
B	**14-uniform**	**65.19%**	**0.6586**

Table 2. Headline results. Track A val dir is top-1 on the 6,745-clip official validation set; Track B is internal val / Kaggle public+private top-1.

3.1. Track A ablations

Regularisation warm-up (+36.3 pp). Identical architecture, single variable: without warm-up VFL stalls at 9.45% (killed epoch 4) and R(2+1)D [13] at 10.78%; with a 10-epoch warm-up VFL reaches 45.78%. The effect transfers across architectures, indicating a general property of cold-start training with label mixing.

Architecture comparison (TSM vs. VFL). At matched settings (rotating folds, T=4, no TTA), TSM-R18 reaches 39.45% vs. VFL 36.44% (+3.01 pp for TSM). This aligns with our architectural analysis: SSv2 classes are defined by *local* motion trajectories (a single finger, an object edge), and TSM mixes time at every spatial location before pooling, while VFL only sees motion in the globally pooled frame vectors. On the classes where VFL has an advantage (global before/after comparison, e.g. *Folding*↔*Unfolding*), VFL's full global attention helps; on directional motion and subtle displacement classes, TSM's local shift wins (Fig. 8).

ResNet50 backbone (+3.4 pp). Replacing ResNet18 with ResNet50 (11.2 → 23.6 M parameters) under the correct T=4 rotating-fold protocol brings 39.45% → 42.85% `val_dir` (+3.40 pp). This confirms the frame-count hypothesis: when temporal inputs are correct, additional backbone capacity translates to better spatial feature extraction, which in turn improves temporal discrimination.

Ablation: focal loss (γ sweep). We ran a controlled focal γ sweep with all other settings frozen (Fig. 4): CE (γ=0): 38.92%; γ=1: 39.50%; γ=2: 39.45%. Any focal weight beats CE by ≈0.5 pp; the difference between γ=1 and γ=2 (0.05 pp) is within noise. For this dataset the optimal γ is in $[1, 2]$ and the exact value is not critical. The CE-trained model (γ=0) reaches 38.92% standalone but contributes +0.52 pp to the 5-model ensemble by providing complementary calibration.

Configuration	Top-1	Δ
Full ensemble (5 models)	**46.75%**	—
− TSM-ResNet50 (rot)	45.41%	−1.33
− VFL-Ultra-rot +TTA	46.11%	−0.64
− TSM-no-focal (rot)	46.23%	−0.52
− TSM-R18-v2-rot +TTA	46.27%	−0.47
− TSM-R18-v2 +TTA	46.43%	−0.31

Table 3. Track A leave-one-out on `val_dir`. Every component contributes; backbone diversity (R50 vs. R18) is the dominant source of gain.

Submission	Kaggle	Δ
Leaky V-JEPA (discarded)	0.81	—
V-JEPA-ft standalone (clean)	0.6361	—
V-JEPA-pseudo standalone	0.6410	+0.5
V-JEPA-heavy (3:1 weight)	0.6499	+1.4
12-uniform	0.6546	+1.9
14-uniform	**0.6586**	+2.3

Table 4. Track B Kaggle submission history (Δ vs. first honest baseline). Diversity drives every gain after the initial fine-tuning.

Ablation: VFL depth and fold count. Reducing VFL from 3 to 2 Transformer layers (Eq. (2): $P_{\text{Transformer}}$ drops from $9.4 \to 6.3$ M) costs $36.44\% \to 35.36\%$ (-1.08 pp): the third layer provides a real but modest gain at this data scale. Increasing the fold count from 5 to 10 (90% data/epoch instead of 80%) gives 35.98% vs. 36.44% (-0.46 pp): VFL has saturated its data-utilisation capacity at k=5 (Fig. 5).

Ensemble and leave-one-out. Table 3 shows the leave-one-out analysis for the final 5-model ensemble. TSM-R50 is by far the dominant contributor (-1.33 pp when removed): backbone diversity (R50 vs. R18) provides more ensemble benefit than same-backbone variants. The CE-trained TSM (-0.52 pp) adds calibration complementarity not captured by the focal-trained models.

3.2. Track B ablations

Kaggle progression. Table 4 traces all clean submissions. The 21 pp gap between the discarded leaky run (0.81, Sec. 3.4) and the first honest baseline (0.6361) directly measures the combined leakage. Subsequent gains come entirely from model diversity.

Diversity beats weighting. A 6-snapshot V-JEPA-only ensemble (0.6410) under-performs the 12-model uniform mix (0.6546), and manual 3:1 V-JEPA weighting (0.6499) falls between them. Uniform averaging wins because architecturally distinct families make uncorrelated errors; two V-JEPA variants agree 0.85 on predictions (Table 5), making a V-JEPA-only ensemble near-redundant.

Family A	Family B	Agreement
V-JEPA-pseudo	TSM	0.370
V-JEPA-ft	TSM	0.372
K400-Base	TSM	0.375
V-JEPA-ft	K400-Base	0.582
V-JEPA-ft	V-JEPA-pseudo	0.854

Table 5. Pairwise inter-family prediction agreement (test set). TSM is the most complementary family, despite being the weakest standalone.

3.3. Failure analysis

Both tracks share the same dominant error patterns, reflecting SSv2's design intent (Figs. 8, 9, 10). *(i) Real vs. pretended.* The worst confusion is *Picking up* predicted as *Pretending to pick up* (56/199 clips, $\text{F1} = 0.09$ in Track A): the early trajectory is identical, and the decisive cue—whether the object actually rises—falls in the withheld future. *(ii) Temporal direction.* *Folding*↔*Unfolding* (24–29 mutual errors) share appearance and differ only in motion direction, which is fragile at four frames. *(iii) Static sink.* *Holding something* absorbs misclassified motion clips (*Moving down*: 30; *Opening*: 28). The best classes ($\text{F1} > 0.54$) are clean single-arc motions (*Pulling left→right*, *Pouring*, *Uncovering*). Class size and per-class F1 correlate only weakly (r=0.24), indicating that difficulty comes from semantic ambiguity, not sample scarcity.

3.4. Data-integrity story

An early extraction sampled frames uniformly over the full source clip (0–100%); training V-JEPA on these yielded 0.81 Kaggle. An audit revealed two leakage vectors: (1) frames from the withheld outcome ($> 60\%$) were reachable; (2) the initial V-JEPA checkpoint had been fine-tuned on SSv2 *with ground-truth labels*, so the backbone had seen test videos. Both runs were discarded and the pipeline rebuilt with the self-supervised `vjepa2-vitl-fpc64-256` checkpoint and a 50% hard frame bound.

4. Conclusion and Future Work

We showed that two data-centric decisions dominate all architectural choices in Track A: a regularisation warm-up (+36.3 pp) and matching frame count to the clip length (+4.4 pp, beating a 2× parameter scale-up at the wrong frame count). Once these are correct, backbone capacity does help: TSM-ResNet50 at T=4 with rotating folds reaches 42.85% val-dir, +3.40 pp over the R18 baseline. The key architectural insight is that TSM's per-block channel shift mixes temporal information *before* spatial pooling, giving it access to fine-grained motion cues that VFL's post-pooling global attention misses; the two architectures are nonetheless complementary in an ensemble. The final five-model ensemble reaches **46.75%** `val_dir`.

Track B shows that progressive unfreezing with LLRD enables effective fine-tuning of large self-supervised video transformers (0.6586 Kaggle), and that uniform averaging of architecturally diverse families consistently beats single-family or manually weighted ensembles.

Future work. (1) Running TTA for TSM-R50 (expected $+2$ pp, currently impractical due to inference time); (2) targeted rebalancing for *holding/pretending* sinks; (3) a 16-frame window-capped VideoMAE-Large checkpoint as a 15th ensemble member; (4) applying the T=4 fix and rotating-folds protocol to the Track B TSM family to improve ensemble diversity.

Contributions

Andrea Signoretti designed the full Track A pipeline: TSM-ResNet and VideoFormer-Lite architectures, the Hydra training framework, the regularisation warm-up and frame-count analysis, the rotating k-fold training, the log-softmax ensemble and TTA code, the ResNet50 experiments, and the ablation study. **Florian Guillaumey** designed the full Track B pipeline: window-capped source-frame re-extraction, V-JEPA 2 fine-tuning with LLRD, pseudo-labelling, the 14-model cross-preprocessing ensemble, and the data-integrity audit. Both authors jointly analysed results and wrote this report.

Figures

All figures are referenced from the main text (Secs. 2.1–3.3).

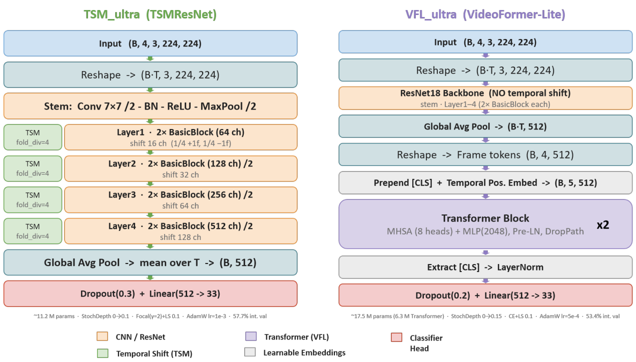

Figure 1. The two Track A temporal architectures. **TSM** mixes time at every residual block, before spatial pooling, via the zero-parameter channel shift of Eq. (1). **VFL** pools each frame first, then applies global self-attention over $T+1$ tokens. Their complementary inductive biases drive the ensemble gain.

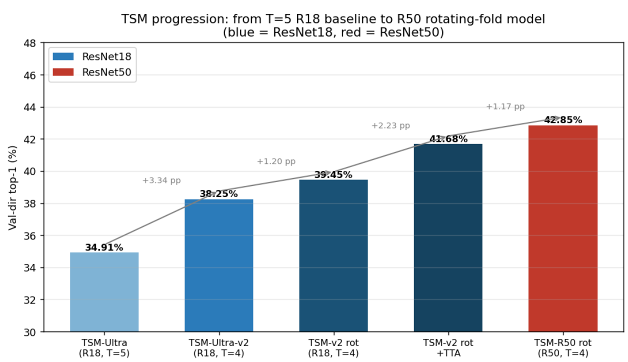

Figure 2. TSM progression: from the T=5 R18 baseline to the R50 rotating model. Correcting the frame count (+3.34 pp) and then scaling the backbone (+3.40 pp) are the two largest single steps. The T=8 R50 scale-up *regresses* because duplicated frames corrupt the TSM shift.

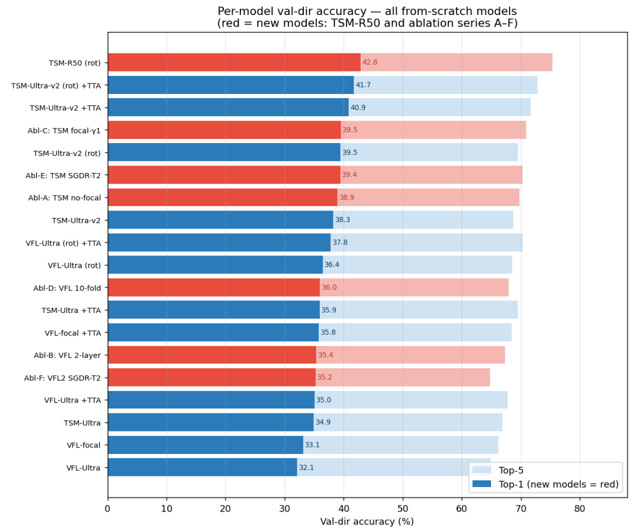

Figure 3. Per-model `val_dir` top-1/top-5 (Track A, all from scratch). Red bars: new models (TSM-ResNet50 and ablation series A–F). The R50 rotating model is the new best single model at 42.85%.

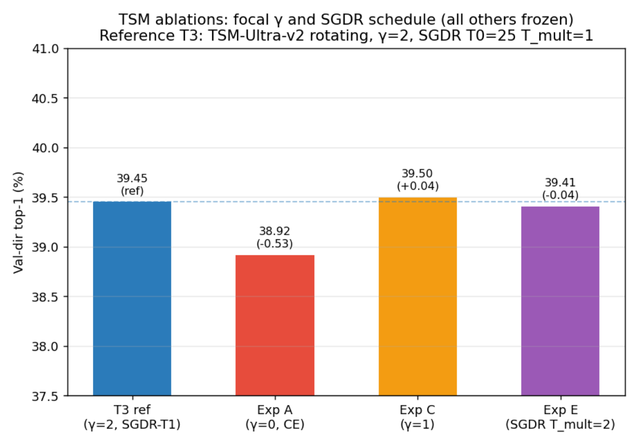

Figure 4. TSM ablation: focal γ sweep and SGDR schedule shape (all other settings frozen). Any focal weight beats CE by $\approx$0.5 pp; the difference between γ=1 and γ=2 is within noise. SGDR T_{mult} has negligible effect (< 0.05 pp).

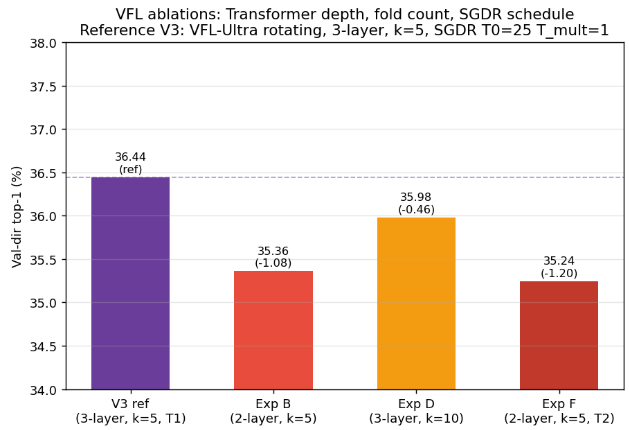

Figure 5. VFL ablations: Transformer depth (2 vs. 3 layers, -1.08 pp for 2 layers), fold count (5 vs. 10, -0.46 pp for 10 folds), and SGDR schedule shape (≤ 0.12 pp effect). VFL is data-saturated at $k{=}5$ and benefits from a 3rd Transformer block.

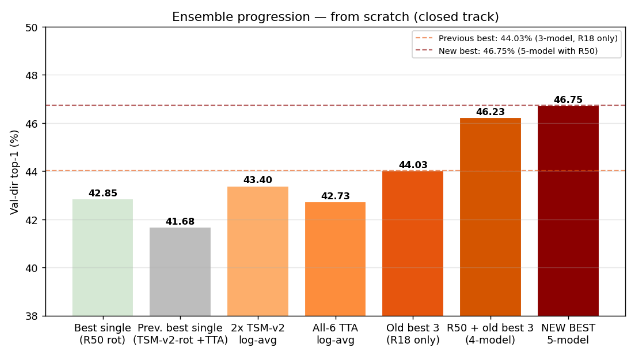

Figure 6. Track A ensemble progression. Adding TSM-R50 to the previous 3-model R18 ensemble raises 44.03% to 46.23%; adding the CE-trained TSM ablation reaches **46.75%**. The R50 backbone is the dominant component (-1.33 pp LOO).

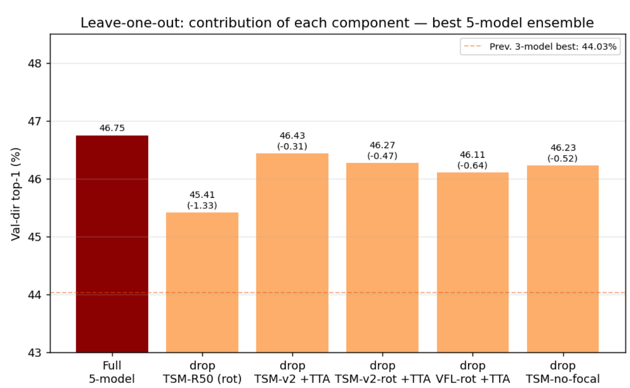

Figure 7. Leave-one-out on the 5-model ensemble. The previous 3-model best (44.03%) is the dashed reference. TSM-R50 contributes -1.33 pp alone.

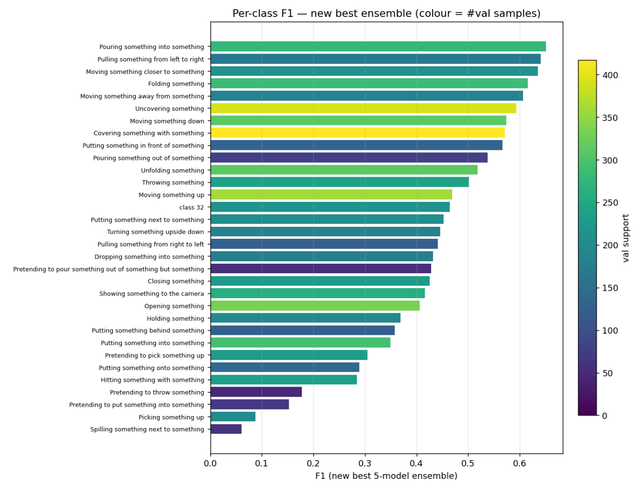

Figure 8. Per-class F1 (5-model best ensemble, `val_dir`). Directional single-object motions score highest; "pretending" and direction-ambiguous classes score lowest.

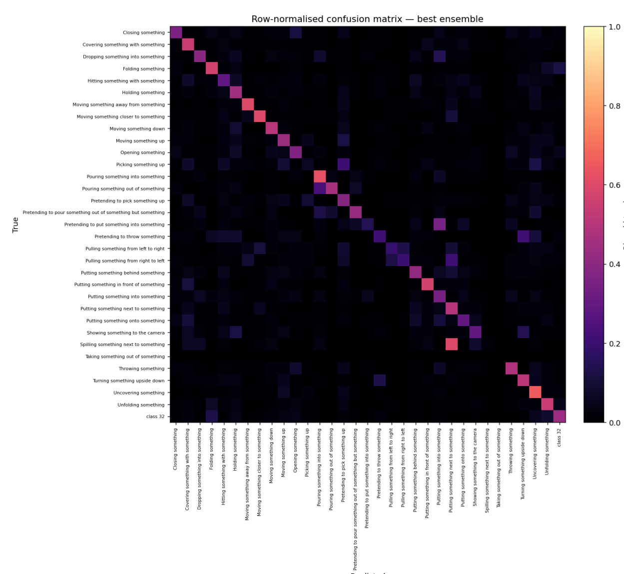

Figure 9. 33×33 confusion matrix (best ensemble, `val_dir`). Off-diagonal mass concentrates on semantically adjacent pairs (real vs. pretended; fold vs. unfold; motion vs. *holding*).

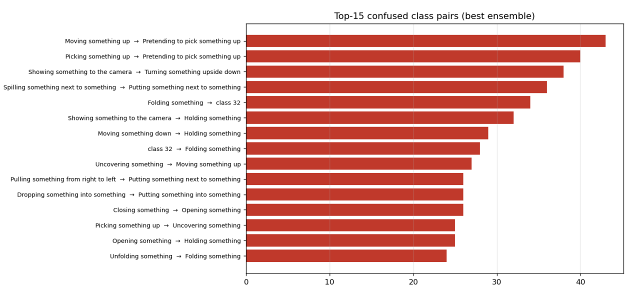

Figure 10. Top-15 confused class pairs. Intent (real vs. pretended) and temporal direction dominate the error distribution.

References

[1] Adrien Bardes, Quentin Garrido, Jean Ponce, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. *arXiv preprint arXiv:2506.09985*, 2025. 1, 3

[2] Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. RandAugment: Practical Automated Data Augmentation with a Reduced Search Space. In *CVPR Workshops*, 2020. 2

[3] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The "Something Something" Video Database for Learning and Evaluating Visual Common Sense. In *ICCV*, 2017. 1

[4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. In *CVPR*, 2016. 1, 2

[5] Gao Huang, Yu Sun, Zhuang Liu, Daniel Sedra, and Kilian Q. Weinberger. Deep Networks with Stochastic Depth. In *ECCV*, 2016. 2

[6] Will Kay, João Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, et al. The Kinetics Human Action Video Dataset. *arXiv preprint arXiv:1705.06950*, 2017. 3

[7] Ji Lin, Chuang Gan, and Song Han. TSM: Temporal Shift Module for Efficient Video Understanding. In *ICCV*, 2019. 1, 3

[8] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal Loss for Dense Object Detection. In *ICCV*, 2017. 2

[9] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic Gradient Descent with Warm Restarts. In *ICLR*, 2017. 2

[10] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *ICLR*, 2019. 2

[11] Rafael Müller, Simon Kornblith, and Geoffrey Hinton. When Does Label Smoothing Help? In *NeurIPS*, 2019. 2

[12] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training. In *NeurIPS*, 2022. 1, 3

[13] Du Tran, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri. A Closer Look at Spatiotemporal Convolutions for Action Recognition. In *CVPR*, 2018. 3

[14] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On Layer Normalization in the Transformer Architecture. In *ICML*, 2020. 2

[15] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. CutMix: Regularization Strategy to Train Strong Classifiers with Localizable Features. In *ICCV*, 2019. 2

[16] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond Empirical Risk Minimization. In *ICLR*, 2018. 2